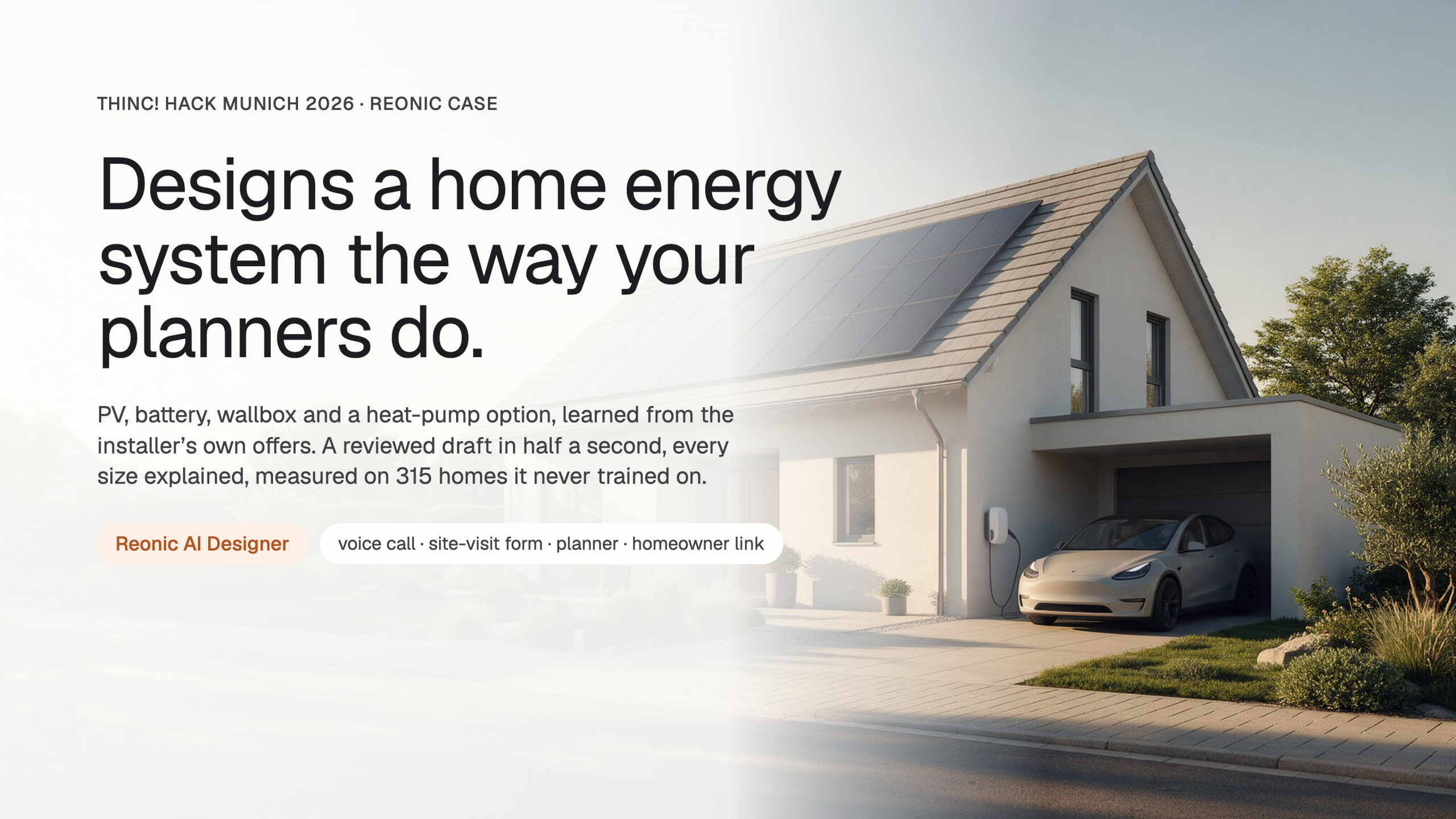
THINC! HACK MUNICH 2026 · REONIC CASE

Designs a home energy system the way your planners do.

PV, battery, wallbox and a heat-pump option, learned from the installer's own offers. A reviewed draft in half a second, every size explained, measured on 315 homes it never trained on.

Reonic AI Designer

voice call · site-visit form · planner · homeowner link



THE PROBLEM

Planners “spend hours on every single design”, and “the first credible offer often wins”.

Most of a design is the same judgement every time: how much PV for this consumption, whether a battery pays, whether a wallbox belongs in the offer. That judgement already exists in the installer’s past offers. Nobody is using it.

The planner

Several offers a day between site visits. Wants a solid first design now, keeps the final say, fears a black box.

The homeowner

Has an offer, has to sign. Wants plain language and room to try variants, fears being oversold.

Quotes: the Reonic challenge brief, pages 3 and 5 · Reonic AI Designer · 2

WHAT WE BUILT

One engine, three doors: the phone call, the planner, the homeowner.

1 · CALL OR PHONE FORM

An ElevenLabs agent takes the site-visit call; every answer lands on screen while he talks.

2 · PLANNER

The real house from the satellite, panels on its roof, every size with a range and its reasons. Lock or type a change.

3 · HOMEOWNER

One link: their house, the system, plain words to read or listen to, and only the choices the installer allows.



The engine decides every size. Claude only turns the planner's words into input changes and engine facts into explanations.

Screenshot: our planner in production · Next.js on Vercel · Python engine on Railway · Supabase · ElevenLabs · Claude · Reonic AI Designer · 3

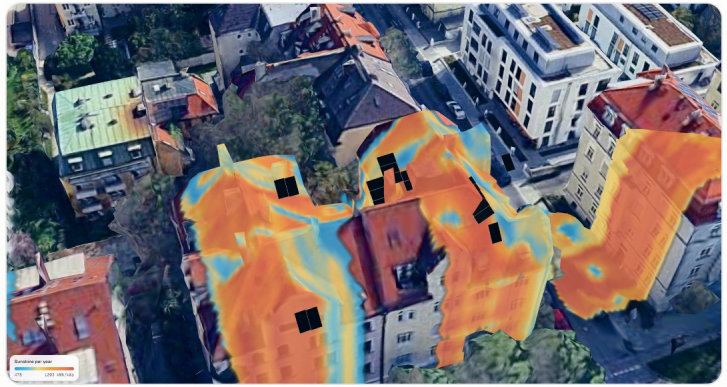
BUILT FOR THE PLANNER'S HANDS

The customer's own house on screen, and every part of it answers to the planner.



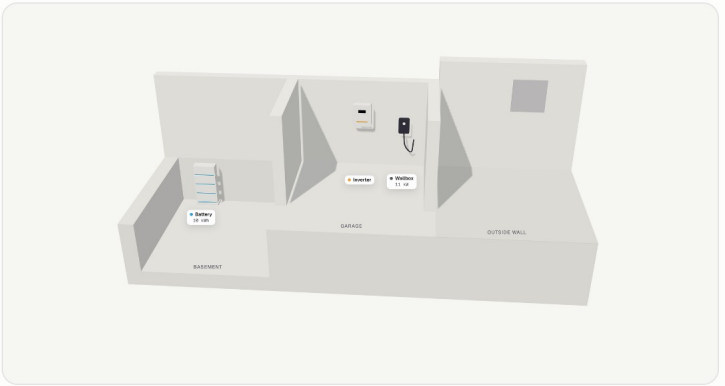
Their house, their street

Type the address: Google Solar fills the roof's kWp, tilt and orientation, and the real street loads around the 3D model.



Panels where the sun is

The sun map shows why the panels sit where they sit. Drag, add or remove them; the engine's kWp never changes.



Inside the house

Upload a floor plan: battery, inverter and wallbox snap into their rooms. Click any part and the focus panel explains the decision.

"Solar already on the roof?" is always the first question, in the call, the site-visit form and the planner: it is the biggest single error source in the data. And the installer decides which variants the homeowner may change; the server rejects everything else.

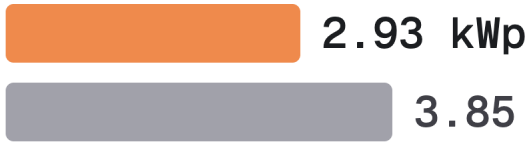
DOES IT DECIDE LIKE AN EXPERT?

Measured on 315 homes the engine never saw.

Held out by customer, never trained on. Every metric is compared with the best of three rules of thumb: kWh ÷ 1,000, the installer's median per consumption band, the installer's kWp per MWh. All homes, without the "solar already on the roof?" answer.

Whole-project solar error, all 315 homes

An unwanted or missing system counts in full. Lower is better.



24 % lower than the best rule

Add-on sales: proposes no new solar where none was sold

62 homes that bought a battery or wallbox for an existing system. Higher is better.



Rules of thumb propose solar for almost every home

PV yes/no right, all homes

New system or add-on, recognised from the planner's inputs. Higher is better.



Better than the best rule

Engine Best rule of thumb for that metric

GET /eval on the live engine, same numbers on the Reality check screen · Reonic AI Designer · 5

THE HONEST HALF, AND THE KNOWN LIMITS

Slightly better on new systems; honest where it is not.

Metric (new systems, 253 homes, unless noted)	Engine	Best rule	Verdict
PV size error per home	2.45 kWp	2.64 kWp	Slightly better, small margin
PV within ±1.5 kWp of the expert's first option (all homes)	40 %	43 %	Below the rule
Expert's PV inside our P10–P90 range	82 %	target 80 %	Calibrated
Expert's PV within 1.5 kWp of one of our three options	66 %	40–45 % single	Alternatives earn their place
Battery size error where both chose one (all homes)	2.42 kWh	2.71 kWh	Better
Battery within one catalogue size (all homes)	56 %	51 %	Better
Trained on one installer, tested on the other (PV size, both ways)	2.95 · 2.82	3.03 · 3.12	Better, small margin

Known limits: wallbox yes/no 72 vs 73 %, no EV data · battery yes/no 88 vs 90 % · top-3 down 3 points with every-option sizes, 69 → 66 % · 450 W per panel assumed for installer A · two installers · placeholder prices · minutes saved and win rate not measured.

NOT A BLACK BOX

The planner can justify every number to the homeowner.

8.1–10.4 kWp

A calibrated range

The expert's PV falls inside the P10–P90 range for 82 % of new full systems. Target: 80 %.

+1.0 kWp

Reasons, from the model itself

Per-decision feature attributions become sentences: "heating demand +1.0 kWp, the recent offer trend +0.7". Model confidence is called what it is.

5 homes

Similar past projects

The five closest homes this installer already built, with what they got. Real history, not an argument.

7.65 · 9.45 · 10.8

One smaller, one larger

Alternatives are real sizes this installer sells, each with its share. The expert's PV is within 1.5 kWp of one of our three options for 66 % of new systems the engine never saw.

Larger than 99 %

Every override is validated

A locked size is compared with the 40 most similar past homes and with physics thresholds calibrated on what these experts actually sold. The brief's second use case, design validation.

Same in, same out

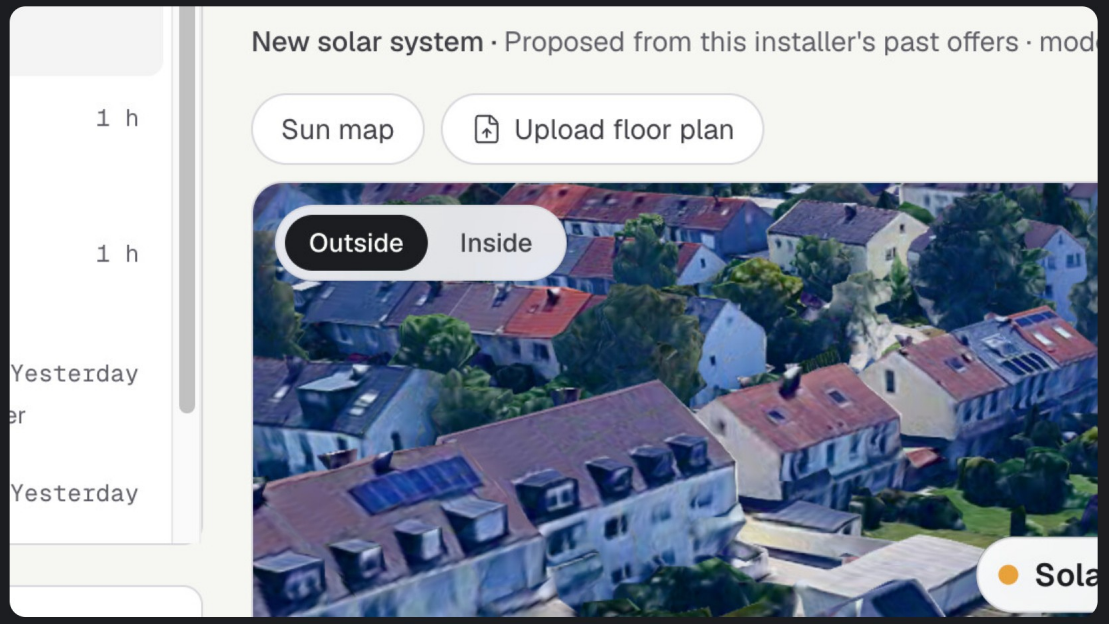
Deterministic and honest

Fixed seeds: two trainings give identical designs. When the engine is unreachable the app shows "Offline estimate" instead of pretending.

Numbers: docs/pitch-facts.md §6 and engine/RESULTS.md round 3 · Reonic AI Designer · 7

THE HOUSE IN 3D: TODAY AND NEXT

Their real street today; drone-sharp detail next.



Today, live in the product

Google's 3D city model of the actual address, with panels placed on the real roof faces. Current limit: the public 3D data is coarse at roof level.



Next: high-resolution capture (concept)

The same pipeline on a drone or high-res photogrammetry scan, like the one Reonic already offers: same engine, same panels, sharp enough to measure every obstacle.

VALUE, MEASURED, NOTHING INVENTED

A reviewed draft in seconds instead of a design from scratch.

29 s

from the start of a scripted call to the homeowner link, 6 clicks. 24.5 s of that is the call itself.

4.6 s

from the end of the call to a shared offer, including a plain-words change by the planner.

0.5 s

for a full design from the engine. Machine time; the planner's own review comes on top.

For the installer

Faster offers from a draft the planner trusts and can override. Consistent designs: same inputs, same design, so junior planners design like the installer's own history. Unusual choices in an offer are flagged against that history before it goes out.

For the homeowner

Their house, their system and the reasons in plain words, to read or to listen to. Variants only within what the installer allows; the home's facts stay fixed. Less fear of being oversold, because the sizes come from homes like theirs.

For Reonic

Three use cases, one engine: planner draft, design validation, homeowner exploration, plus voice intake. It learns per installer from data Reonic already holds, and tells when it is only on par.

Timings measured on production, 8 Nov 2026 (docs/pitch-facts.md §8) · Reonic AI Designer · 9

THE FIVE-MINUTE STORY YOU JUST SAW

1
A call: the house builds while he talks, EV included

2
A complete system, with the reasons

3
“As independent as possible”: the design adapts

4
Checked against the expert, on 315 unseen homes

5
One link for the homeowner

One engine for the planner, the homeowner and the phone call. Every size explained. 24 % less solar error than the best rule of thumb, on homes it never saw.

Reonic AI Designer Code, README and evaluation in the repository · demo on production · thinc! Hack Munich 2026

HOW IT DECIDES

One chain, not three guesses: each decision sees the one before it.

Heat pump

A rule, offered as a customer option for gas and oil homes. Heat demand ÷ 2,450 h, snapped to the sizes the installer sold. Never imposed.

→

PV yes/no, then size

Sees the total demand incl. EV and heat pump. Gradient-boosted trees with a robust loss; more demand or an EV can never shrink the PV. P10–P90 range, conformally calibrated.

→

Battery yes/no, then size

Sees the planned PV. Size = the classifier's pick blended with the installer's own kWh-per-kWp rule, snapped to its catalogue.

→

Wallbox

EV → wallbox. The simple rule beat the model on training data, so the engine uses the rule and says so.

Before the chain: new system or add-on?

Battery retrofit, wallbox-only or new PV is its own decision. The question "Solar already on the roof?" is its strongest input, so the call and the form ask it first.

Learned from

1,994 offer options across 1,593 signed projects from two installers, parsed from the offers' line items, 20 projects hand-checked. Decisions learned from each project's first full design, sizes from every option: Reonic confirmed each option is a valid design. Every size snaps to the installer's catalogue.

Physics on top

An hourly year with PVGIS yield per roof tilt and orientation gives yield, self-consumption and autarky, and flags what is unusual for these installers.

WHAT THE DATA TAUGHT US

Four things experts do that no textbook rule says.

new vs. add-on

The decision that matters is not the kWp

Recognising battery retrofits and wallbox-only orders from the planner's inputs cuts the whole-project solar error by 24 %. Pure size error improves only from 2.64 to 2.45 kWp.

1 input, 1 question

Yearly consumption does most of the work

Installer + consumption alone already reach the band-median rule; removing consumption costs 0.9 kWp. The question "solar already on the roof?" would be worth the most: as a what-if, 2.93 → 2.15 kWp over all homes.

2.77 vs 2.49 kWp

Experts do not size for maximum return

Sizing for the economic optimum is worse than learning from them. Installers size PV at about 2 kWp per MWh, and the battery by the PV size, not by consumption: 2.25 vs 4.46 kWh error for the "1 kWh per 1,000 kWh" rule.

7–8 kWp + battery

Small homes get a full system on purpose

New systems for homes using 1,500–2,500 kWh got a median 7.2 kWp from the experts (n = 109). Textbook plausibility rules would flag that as oversized, so our checks are calibrated on expert practice, and the engine proposes the same.

IN THREE NUMBERS

0.5 s

A complete design

PV, battery, wallbox, heat-pump option. Every size with its reasons, range and alternatives.

24 %

Less solar error than the best rule of thumb

Because it tells a new system from an add-on, the way the installer's own planners do.

315

Homes it never saw

Held out by customer. Every number in this deck is measured there, including where we are only on par.

One engine behind a phone call, a planner workspace and a homeowner link. The engine decides every size; an LLM never picks one.